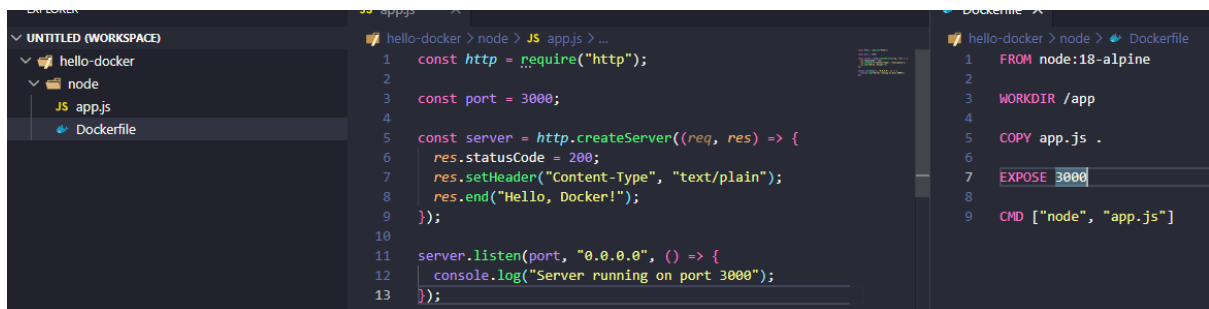


Node



```
hello-docker > node > JS app.js > ...
1  const http = require("http");
2
3  const port = 3000;
4
5  const server = http.createServer((req, res) => {
6    res.statusCode = 200;
7    res.setHeader("Content-Type", "text/plain");
8    res.end("Hello, Docker!");
9  });
10
11  server.listen(port, "0.0.0.0", () => {
12    console.log("Server running on port 3000");
13  });
```

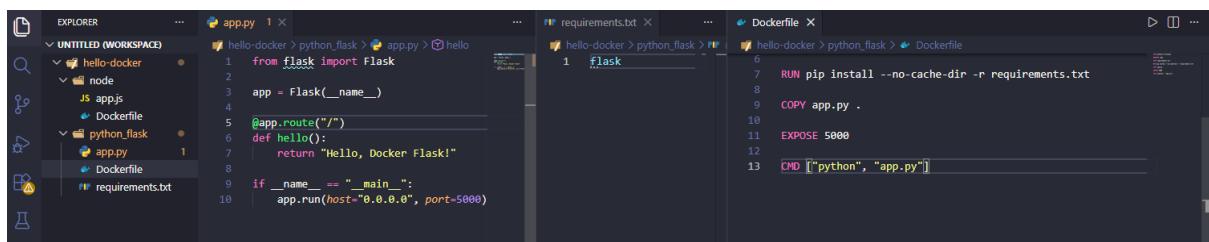
```
hello-docker > node > Dockerfile
1  FROM node:18-alpine
2
3  WORKDIR /app
4
5  COPY app.js .
6
7  EXPOSE 3000
8
9  CMD ["node", "app.js"]
```

```
PS T:\Kiến trúc máy tính\Thực hành\Tuan04\hello-docker\node> docker build -t hello-docker-node .
[+] Building 179.9s (9/9) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 130B
```

```
View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/z4ka7nj837e97bq2efk7bnn58
PS T:\Kiến trúc máy tính\Thực hành\Tuan04\hello-docker\node> docker run -p 3000:3000 hello-docker-node
Server running on port 3000
```



Python flask



```
hello-docker > python_flask > app.py > hello
1  from flask import Flask
2
3  app = Flask(__name__)
4
5  @app.route("/")
6  def hello():
7    return "Hello, Docker Flask!"
8
9  if __name__ == "__main__":
10  app.run(host="0.0.0.0", port=5000)
```

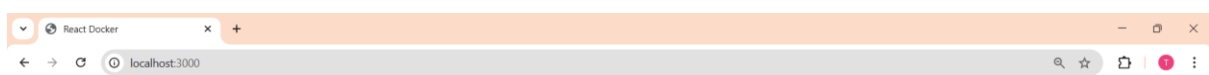
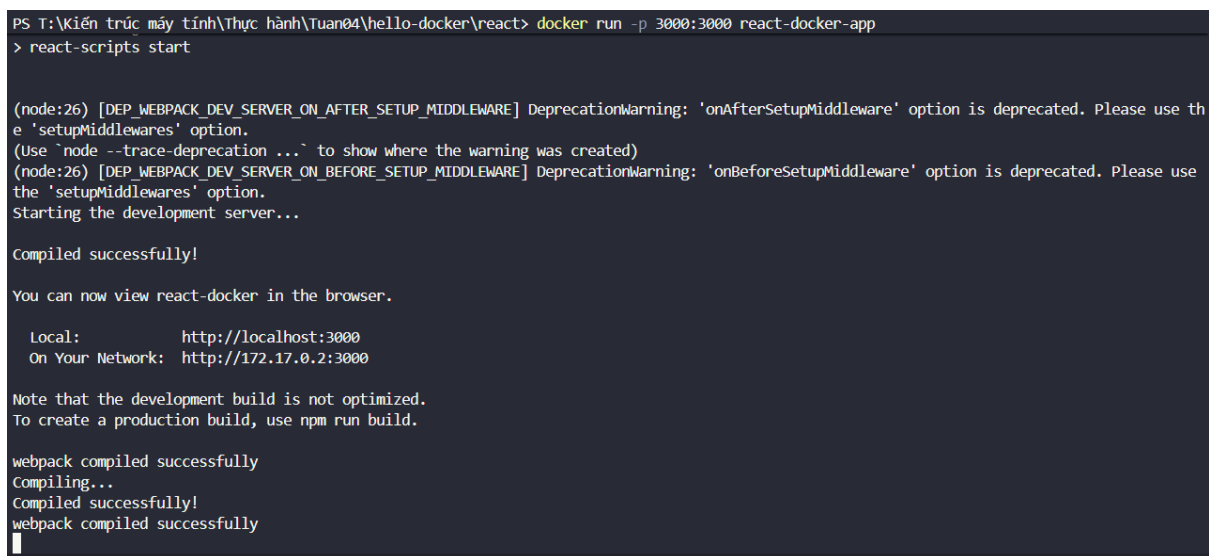
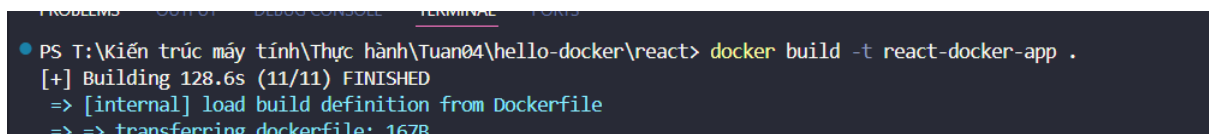
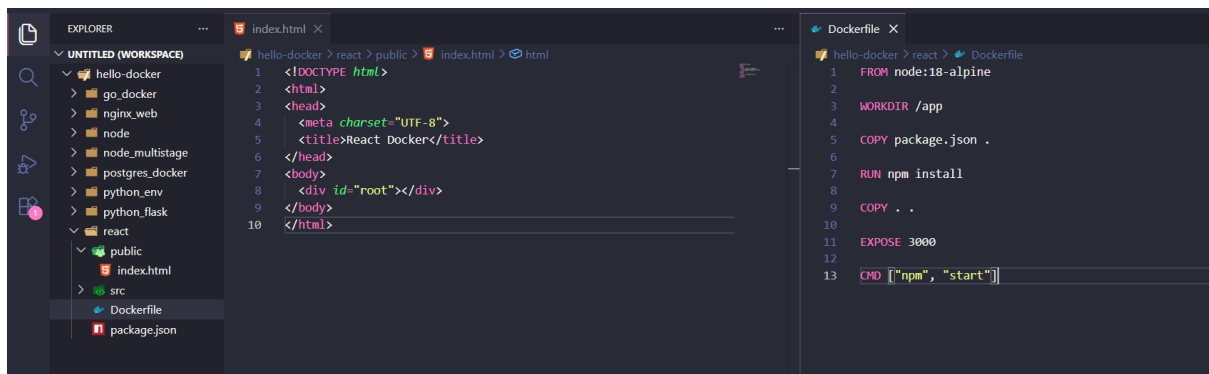
```
hello-docker > python_flask > Dockerfile
6
7  RUN pip install --no-cache-dir -r requirements.txt
8
9  COPY app.py .
10
11  EXPOSE 5000
12
13  CMD ["python", "app.py"]
```

```
Error: failed to build: failed to solve: canceled: context canceled
PS T:\Kiến trúc máy tính\Thực hành\Tuan04\hello-docker\python_flask> docker build -t python_flask .
[+] Building 89.6s (10/10) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 218B
=> [internal] load metadata for docker.io/library/python:3.9-alpine
=> [internal] load .dockerignore
```

```
Press CTRL+C to quit
PS T:\Kiến trúc máy tính\Thực hành\Tuan04\hello-docker\python_flask> docker run -p 5000:5000 python_flask
* Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.17.0.3:5000
Press CTRL+C to quit
172.17.0.1 - - [13/Mar/2026 15:17:09] "GET / HTTP/1.1" 200 -
172.17.0.1 - - [13/Mar/2026 15:17:09] "GET /favicon.ico HTTP/1.1" 404 -
```

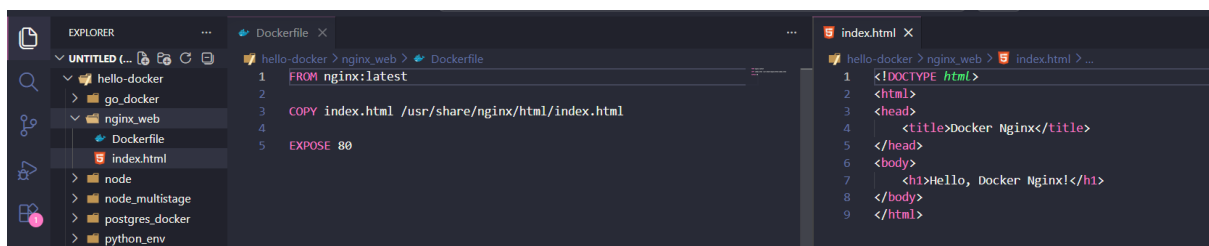


React



Hello, Docker React!

nginx_web



```

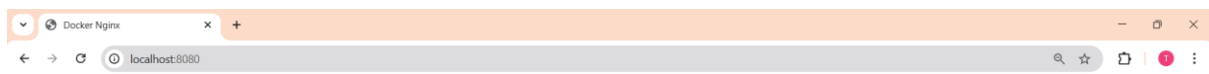
PS T:\Kiến trúc máy tính\Thực hành\Tuan04\hello-docker\nginx_web> docker build -t nginx-web .
[+] Building 9.9s (5/7)
PS T:\Kiến trúc máy tính\Thực hành\Tuan04\hello-docker\nginx_web> docker build -t nginx-web .
PS T:\Kiến trúc máy tính\Thực hành\Tuan04\hello-docker\nginx_web> docker build -t nginx-web .
[+] Building 18.2s (8/8) FINISHED
=> [internal] load build definition from Dockerfile

```

```

PS T:\Kiến trúc máy tính\Thực hành\Tuan04\hello-docker\nginx_web> docker run -p 8080:80 nginx-web
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/03/14 12:36:26 [notice] 1#1: using the "epoll" event method
2026/03/14 12:36:26 [notice] 1#1: nginx/1.29.6
2026/03/14 12:36:26 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/03/14 12:36:26 [notice] 1#1: OS: Linux 6.6.87.2-microsoft-standard-WSL2
2026/03/14 12:36:26 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/03/14 12:36:26 [notice] 1#1: start worker processes
2026/03/14 12:36:26 [notice] 1#1: start worker process 29
2026/03/14 12:36:26 [notice] 1#1: start worker process 30
2026/03/14 12:36:26 [notice] 1#1: start worker process 31
2026/03/14 12:36:26 [notice] 1#1: start worker process 32
2026/03/14 12:36:26 [notice] 1#1: start worker process 33
2026/03/14 12:36:26 [notice] 1#1: start worker process 34
2026/03/14 12:36:26 [notice] 1#1: start worker process 35
2026/03/14 12:36:26 [notice] 1#1: start worker process 36
2026/03/14 12:36:26 [notice] 1#1: start worker process 37
2026/03/14 12:36:26 [notice] 1#1: start worker process 38
2026/03/14 12:36:26 [notice] 1#1: start worker process 39
2026/03/14 12:36:26 [notice] 1#1: start worker process 40

```



Hello, Docker Nginx!

Go



```
linux/r3qoxpxuk9gtxm53mnqrx12k6 (ctrl + click)
View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/f3qoxpxuk9gtxm53mnqrx12k6
PS T:\Kiến trúc máy tính\Thực hành\Tuan04\hello-docker\go_docker> docker run -p 8080:8080 go-docker-app
Server running on port 8080

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/eie0gbh4jqeugd2d2cnxslzr6
PS T:\Kiến trúc máy tính\Thực hành\Tuan04\hello-docker\go_docker> docker build -t go-docker-app .
[+] Building 14.5s (10/10) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 172B
=> [internal] load metadata for docker.io/library/golang:1.21-alpine
=> [internal] load .dockerignore
```



Multi-stage

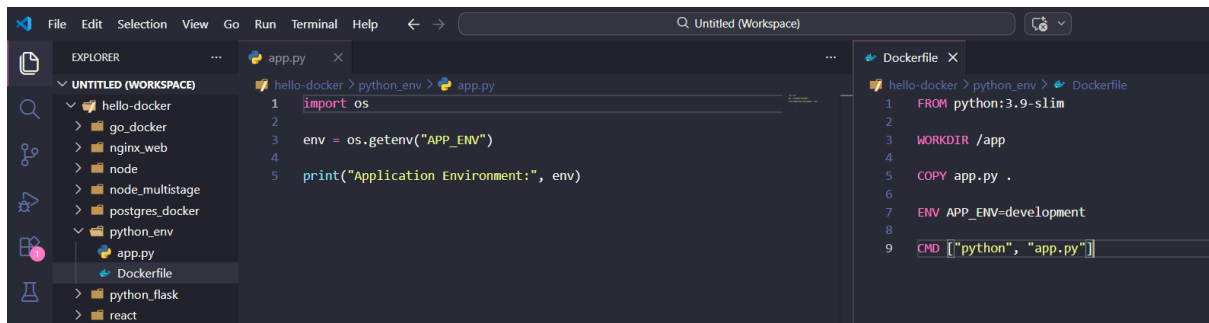
A screenshot of the Visual Studio Code editor. The Explorer sidebar on the left shows a project structure with folders like 'hello-docker', 'node_multistage', and 'react'. The main editor area has three files open: 'app.js', 'Dockerfile', and 'package.json'.
The 'app.js' file contains a simple HTTP server using Node.js and Express, listening on port 3000.
The 'Dockerfile' file is configured for a multi-stage build. Stage 1, 'Build', uses 'node:18 AS builder' to copy 'package.json' and 'app.js', run 'npm install', and copy the application files. Stage 2, 'Runtime', uses 'node:18-alpine' as the base image, copies the built application from the first stage, exposes port 3000, and sets the command to run the application.
The 'package.json' file defines the project as 'node-multistage' with version '1.0.0' and 'app.js' as the main entry point.

```
PS T:\Kiến trúc máy tính\Thực hành\Tuan04\hello-docker\node_multistage> docker build -t node-multistage .
[+] Building 25.3s (8/14)
[+] Building 25.5s (8/14)
[+] Building 29.1s (8/14)
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [builder 1/5] FROM docker.io/library/node:18@sha256:c6ae79e38498325db67193d391e6ec1d224d96c693a8a4d943498556716d3783
=> => resolve docker.io/library/node:18@sha256:c6ae79e38498325db67193d391e6ec1d224d96c693a8a4d943498556716d3783
docker:
docker:
docker:
```

```
View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/dcn3iy16kof2qwtk90muead43
PS T:\Kiến trúc máy tính\Thực hành\Tuan04\hello-docker\node_multistage> docker run -p 3000:3000 node-multistage
Server running on port 3000
```



python_env



```
PS T:\Kiến trúc máy tính\Thực hành\Tuan04\hello-docker\python_env> docker build -t python-env-app .
[+] Building 18.2s (9/9) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 145B
=> [internal] load metadata for docker.io/library/python:3.9-slim
```

```
View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/qgjoa5yxeawhghlubbpulr2wf
PS T:\Kiến trúc máy tính\Thực hành\Tuan04\hello-docker\python_env> docker run python-env-app
Application Environment: development
PS T:\Kiến trúc máy tính\Thực hành\Tuan04\hello-docker\python_env>
```

postgres

docker build -t postgres-custom .

docker run -p 5432:5432 postgres-custom